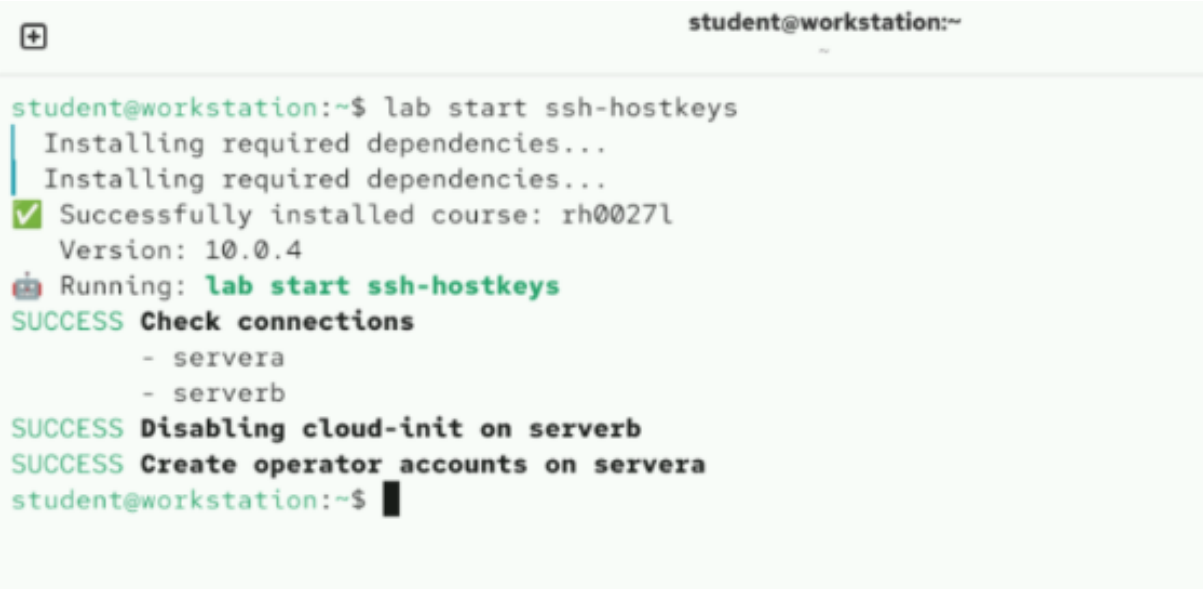
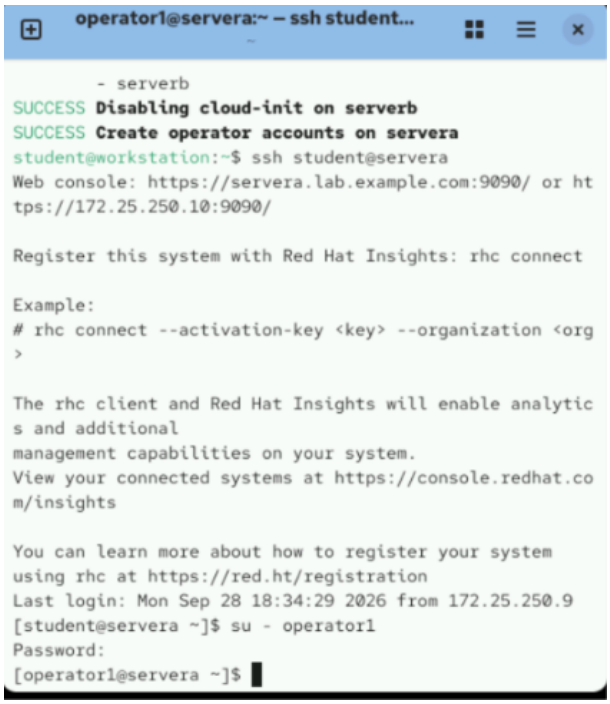
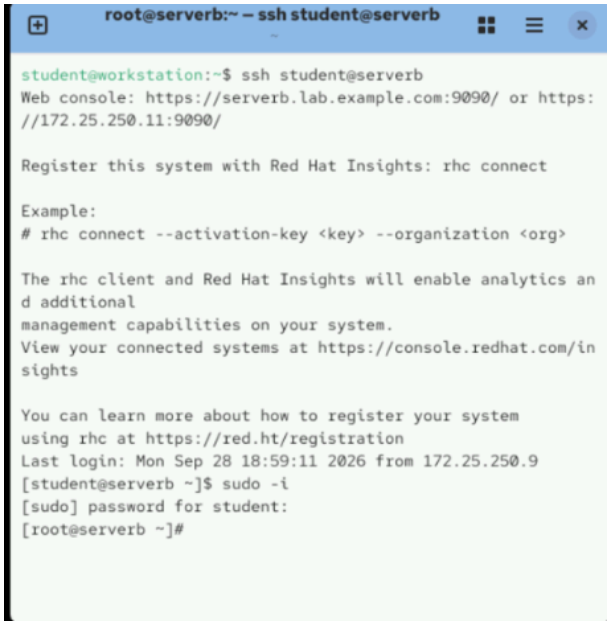
19.2- Guided Exercise: Manage SSH Host Keys

Ahmed Abdelhedi



1/





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```
[operator1@servera ~]$ ls -l ~/.ssh
ls: cannot access '/home/operator1/.ssh': No such file or directory
[operator1@servera ~]$ ssh student@serverb
The authenticity of host 'serverb (172.25.250.11)' can't be established.
ED25519 key fingerprint is SHA256:OYgkbBUEW+9QL1Z2o973G5ZiUpzymP+E5PXP8rnL5s4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? no
Host key verification failed.
[operator1@servera ~]$
```

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```
[root@serverb ~]# ssh-keygen -l -f /etc/ssh/ssh_host_ed25519_key.pub
256 SHA256:OYgkbBUEW+9QL1Z2o973G5ZiUpzymP+E5PXP8rnL5s4 root@serverb (ED25519)
[root@serverb ~]#
```

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```
[operator1@servera ~]$ ssh student@serverb
The authenticity of host 'serverb (172.25.250.11)' can't be
established.
ED25519 key fingerprint is SHA256:OYgkbBUEW+9QL1Z2o973G5ZiUp
zymP+E5PXP8rnL5s4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[finger
print])? yes
Warning: Permanently added 'serverb' (ED25519) to the list o
f known hosts.
student@serverb's password:
Web console: https://serverb.lab.example.com:9090/ or https:
//172.25.250.11:9090/

Register this system with Red Hat Insights: rhc connect

Example:
# rhc connect --activation-key <key> --organization <org>

The rhc client and Red Hat Insights will enable analytics an
d additional
management capabilities on your system.
View your connected systems at https://console.redhat.com/in
sights

You can learn more about how to register your system
using rhc at https://red.ht/registration
Last login: Mon Sep 28 19:34:31 2026 from 172.25.250.9
[student@serverb ~]$
```

```
Last login: Mon Sep 28 19:34:31 2026 from 172.25.250.9
[student@serverb ~]$ exit
logout
Connection to serverb closed.
[operator1@servera ~]$
```

```
[operator1@servera ~]$ ssh-keygen -l -f ~/.ssh/known_hosts
256 SHA256:OYgkbBUEW+9QL1Z2o973G5ZiUpzymP+E5PXP8rnL5s4 serve
rb (ED25519)
3072 SHA256:m8e1kErQJSC0VFwqtex/U0iB1voqCHWwUFw95c4iQ14 serv
erb (RSA)
256 SHA256:DHptoXnrmrwewQPC6lLYYHFw/iWXIuvp/wyGlFynMG4 serve
rb (ECDSA)
[operator1@servera ~]$
```

```
[operator1@servera ~]$ ssh student@serverb 'hostname'
student@serverb's password:
serverb
[operator1@servera ~]$
```

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```
[root@serverb ~]# rm -f /etc/ssh/ssh_host_*
[root@serverb ~]# systemctl restart sshd
[root@serverb ~]# ls -l /etc/ssh/ssh_host_*
-rw-------. 1 root root  492 Sep 28 19:41 /etc/ssh/ssh_host_
ecdsa_key
-rw-r--r--. 1 root root  162 Sep 28 19:41 /etc/ssh/ssh_host_
ecdsa_key.pub
-rw-------. 1 root root  387 Sep 28 19:41 /etc/ssh/ssh_host_
ed25519_key
-rw-r--r--. 1 root root   82 Sep 28 19:41 /etc/ssh/ssh_host_
ed25519_key.pub
-rw-------. 1 root root 2578 Sep 28 19:41 /etc/ssh/ssh_host_
rsa_key
-rw-r--r--. 1 root root  554 Sep 28 19:41 /etc/ssh/ssh_host_
rsa_key.pub
[root@serverb ~]# ssh-keygen -l -f /etc/ssh/ssh_host_ed25519
_key.pub
256 SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU no co
mment (ED25519)
[root@serverb ~]# exit
logout
[student@serverb ~]$ exit
logout
Connection to serverb closed.
student@workstation:~$
```

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```
[operator1@servera ~]$ ssh student@serverb 'hostname'
student@serverb's password:
serverb
[operator1@servera ~]$ ssh student@serverb 'hostname'
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@    WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!     @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-
middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host
is
SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU.
Please contact your system administrator.
Add correct host key in /home/operator1/.ssh/known_hosts to
get rid of this message.
Offending ECDSA key in /home/operator1/.ssh/known_hosts:3
Host key for serverb has changed and you have requested stri
ct checking.
Host key verification failed.
[operator1@servera ~]$ ssh-keygen -R serverb
# Host serverb found: line 1
# Host serverb found: line 2
# Host serverb found: line 3
/home/operator1/.ssh/known_hosts updated.
Original contents retained as /home/operator1/.ssh/known_hos
ts.old
[operator1@servera ~]$ ssh student@serverb 'hostname'
The authenticity of host 'serverb (172.25.250.11)' can't be
established.
```

```
[operator1@servera ~]$ ssh-keygen -R serverb
# Host serverb found: line 1
# Host serverb found: line 2
# Host serverb found: line 3
/home/operator1/.ssh/known_hosts updated.
Original contents retained as /home/operator1/.ssh/known_hos
ts.old
[operator1@servera ~]$ ssh student@serverb 'hostname'
The authenticity of host 'serverb (172.25.250.11)' can't be
established.
ED25519 key fingerprint is SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSn
KQoUQxMfKE0s0b2ZU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[finger
print])? yes
Warning: Permanently added 'serverb' (ED25519) to the list o
f known hosts.
student@serverb's password:
serverb
[operator1@servera ~]$ exit
logout
[student@servera ~]$
```

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```
[student@servera ~]$ ssh-keyscan -t ed25519 serverb
# serverb:22 SSH-2.0-OpenSSH_9.9
serverb ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIG/KewZghU1wIeRIvulS9pJJWVRruk0KAqhvj1Ih/Qmz
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 \
>
> serverb >> /etc/ssh/ssh_known_hosts'
[sudo] password for student:
usage: ssh-keyscan [-46cDHqv] [-f file] [-O option] [-p port] [-T timeout]
                   [-t type] [host | addrlist namelist]
bash: line 3: serverb: command not found
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 \ serverb >> /etc/ssh/ssh_known_hosts'
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 \
>
> sudo bash -c 'ssh-keyscan -t ed25519 \
usage: ssh-keyscan [-46cDHqv] [-f file] [-O option] [-p port] [-T timeout]
                   [-t type] [host | addrlist namelist]
usage: ssh-keyscan [-46cDHqv] [-f file] [-O option] [-p port] [-T timeout]
                   [-t type] [host | addrlist namelist]
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 \ serverb >> /etc/ssh/ssh_known_hosts'
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 serverb >> /etc/ssh/ssh_known_hosts'
[student@servera ~]$ sudo bash -c 'ssh-keyscan -t ed25519 serverb >> /etc/ssh/ssh_known_hosts'
[student@servera ~]$ ssh-keygen -l -f /etc/ssh/ssh_known_hosts
256 SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU serverb (ED25519)
256 SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU serverb (ED25519)
256 SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU serverb (ED25519)
256 SHA256:iS9Q0/bXHOwXKX5zTP10nAIxSnKQoUQxMfKE0s0b2ZU serverb (ED25519)
[student@servera ~]$
```

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```
[student@servera ~]$ su - operator2
Password:
[operator2@servera ~]$ ssh student@serverb 'hostname'
student@serverb's password:
serverb
[operator2@servera ~]$ cat ~/.ssh/known_hosts
cat: /home/operator2/.ssh/known_hosts: No such file or directory
[operator2@servera ~]$
```

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```
[operator2@servera ~]$ exit
logout
[student@servera ~]$ exit
logout
Connection to servera closed.
student@workstation:~$ lab finish ssh-hostkeys
🤖 Running: lab finish ssh-hostkeys
SUCCESS Re-enabing cloud-init on serverb
SUCCESS Delete operator accounts on servera
SUCCESS Deleting system-wide known_hosts file on servera
SUCCESS Clear known hosts for student on workstation
student@workstation:~$
```